J26-IT-430 – Datasets and Models (C1, C2, C3)

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For each component: **which datasets** (with download links), **which models** (method, data, baseline, result) and a **short description** of each model. Results are from the training notebooks run on 19 Oct 2026 (test split, never used for training). Full numbers: each component's results/ folder.

Component	Training notebook	Model code (packages)	App
C1 Charging scheduling	charging_scheduling/notebooks/c1_model_training.ipynb	forecasting/, scheduler/, simulator/	Smart Charge LK (run_c1.bat)
C2 Station siting	station_siting/notebooks/c2_model_training.ipynb	demand/, mcdm/, mclp/	GridSite Planner (run_c2.bat)
C3 Route & charging stop	route_recommendation/notebooks/c3_model_training.ipynb	routing/, energy/, charging_stops/	TerrainRoute (run_c3.bat)

C1 – Personalized, Routine-Aware EV Charging Scheduling (When should each EV charge?)

Datasets

Dataset	What it gives	Link	Used for	Licence
ACN-Data (Caltech, JPL, office001)	66,745 EV charging sessions 2018–2021 with drivers' app inputs (requested kWh, departure)	https://ev.caltech.edu/dataset	1.1b EV demand, 1.2 routines, 1.3 need prediction	free token, site terms
Utrecht office parking (4TU)	55,379 sessions 2020–2024, one office car park	https://doi.org/10.4121/80ef3824-3f5d-4e45-8794-3b8791efbd13.v1	1.2 routines, energy values for the scenario	CC BY-NC-SA 4.0 (non-commercial)
City of Boulder EV sessions	67,599 public sessions 2018–2023 (after removing duplicates)	https://open-data.bouldercolorado.gov/datasets/95992b3938be4622b07f0b05eba95d4c_0/about	extra session data (replaces withdrawn Palo Alto)	city open data
UrbanEV (Shenzhen, Dryad)	1,682 stations / 275 zones, hourly occupancy, energy, prices	https://datadryad.org/dataset/doi:10.5061/dryad.np5hqc04z	station busyness (later)	Dryad (CC0)
RECON-SL (Sri Lanka, LIRNEasia)	15-min smart-meter data of ~2,900 Sri Lankan homes, Oct 2023 – Dec 2024 + surveys	https://ieee-dataport.org/node/14951 (DOI 10.21227/n1dk-q860)	1.1a base-load forecast, scheduler base load	IEEE DataPort terms
PUCSL tariff table (Oct 2025)	TOU periods and LKR/kWh	https://www.pucsl.gov.lk/wp-content/uploads/2025/10/Annex-2-Approved-Tariff-Table.pdf	cost of every plan	public
PUCSL generation report / 2012 demand-curve study	national demand shape (charts)	https://www.pucsl.gov.lk	background only	public

Models

#	Model	Method	Baseline	Result (test)
1.1a	Household base-load forecast	XGBoost, day-ahead, 15-min	same time last week / yesterday	**94.1 %** accuracy (last week 90.2 %, yesterday 94.3 %)
1.1b	EV charging demand forecast	XGBoost, day-ahead, 15-min	same time last week / yesterday	**68.7 %** (last week 54.3 %, yesterday 38.1 %)
1.2	Routine clustering	K-Means (elbow + silhouette)	–	1,016 drivers, k = 2, silhouette 0.47
1.3	Need prediction (departure + kWh)	XGBoost + quantile q10	profile defaults, driver app input	departure error **86 min** (default 102); energy error **3.9 kWh** (default 7.3); q10 coverage 87 %
1.4	Coordinated scheduler	Linear programme (SciPy HiGHS): min peak, then fill low-load slots	B1 uncontrolled, B2 TOU 22:30, B3 TOU + delay, B4 no personal info	peak **−63 %** vs B1, 100 % cars ready; B2 makes a **+46 % rebound peak**; B4 only 49.5 % ready
1.5	EV data simulator	Monte Carlo	–	scenarios / demos only (never accuracy)

Short descriptions

- **1.1a Base-load forecast** – predicts how much electricity an average Sri Lankan home will use in every 15 minutes of tomorrow, using only data at least 24 h old (yesterday, last week, time of day, weekday). Very accurate, but the simple "same time yesterday" is just as good.
- **1.1b EV demand forecast** – the same idea for the total charging power of a group of chargers; harder to predict, here XGBoost is clearly better than the baselines.
- **1.2 Routine clustering** – groups drivers by habits (arrival time, how regular, how long they stay, kWh, weekend use). With workplace data only 2 weak groups appear; home routines need Sri Lankan pilot data.
- **1.3 Need prediction** – at plug-in, predicts when the car will leave and how much energy it needs, from the driver's own past sessions. A "safe early" (q10) departure keeps the car ready. New drivers (< 10 sessions) use profile defaults (cold start).
- **1.4 Coordinated scheduler** – decides how much power each car gets in each 15-min slot so the street's peak stays as low as possible while every car is ready on time. Shows why "everyone at 22:30" is bad (rebound peak).
- **1.5 Simulator** – creates example home-charging evenings for "what if" scenarios.

C2 – Grid-Aware Temporal MCDM Optimization of Charging Station Locations (Where to build?)

Datasets

Dataset	What it gives	Link	Used for	Licence
OpenStreetMap Sri Lanka (Geofabrik)	roads, POIs (fuel, malls, supermarkets, car parks, hotels, offices …), land use	https://download.geofabrik.de/asia/sri-lanka.html	demand index, candidate sites, road access	ODbL – attribution
WorldPop 2020 (100 m)	population per cell	https://data.worldpop.org/GIS/Population/Global_2000_2020/2020/LKA/lka_ppp_2020.tif (portal: https://hub.worldpop.org)	demand index	CC BY 4.0
ESA WorldCover 2021 (10 m)	land cover	https://esa-worldcover.org/en	land suitability	CC BY 4.0
HDX admin boundaries (cod-ab-lka)	district / DS / GN polygons	https://data.humdata.org/dataset/cod-ab-lka	study area (Colombo District)	CC BY-IGO
CEB Long Term Transmission Development Plan 2013–2022	grid substation capacity (MVA) + forecast load to 2025	https://www.pucsl.gov.lk/wp-content/uploads/2019/09/Transmission-Plan-2013-2022.pdf	**substation capacity constraint**	public
OSM power + chargers (Overpass extract)	substation locations (141), existing chargers (66)	https://overpass-api.de	grid distance, existing chargers	ODbL
DMT Statistical Abstract 2024, Table 7.3	new vehicles by fuel 2019–2023	https://www.statistics.gov.lk/abstract2024/CHAP7/7.3.pdf	EV growth scenarios	public
RDA traffic counts · current CEB substation loading · operator charger lists	–	**request via supervisor**	traffic, grid, validation	–

Models

#	Model	Method	Baseline	Result (test / Colombo District)
2.0	Hexagon grid	H3 resolution 8 (~0.78 km²)	–	866 hexagons
2.1	Temporal demand index	weighted sum of population, POIs, road traffic proxy, land use per period	static demand (population only)	4 periods + time-weighted mix
2.2	Hotspots	K-Means vs DBSCAN	–	K-Means k = 3, silhouette 0.41; DBSCAN finds 1 continuous zone
2.3	Criteria weights	AHP (eigenvector, consistency ratio)	equal weights	demand 40 %, grid 24 %, road 14 %, competition 14 %, land 8 %; **CR 0.007**
2.4	Site scoring	weighted-sum MCDM on 1,129 real buildable sites	static MCDM	ranked sites
2.5	Site selection	**MCLP with substation capacity** (MILP, SciPy HiGHS)	traffic-only top-N, K-Means centres, static MCDM, MCLP without grid	**84.3 % demand covered, 0 overloaded substations** (no-grid 85.3 % / 1 overload; K-Means centres 77.9 %; static MCDM 18.4 %; traffic-only 14.8 %)
Val.	Validation	grid stress test, hold-out, ±20 % weights	–	stays at 0 overloads up to 6 MW stations (no-grid: 8 overloads); hold-out hit rate 50 % vs 14 %; top-10 overlap ≥ 90 %, Kendall τ ≥ 0.96

Short descriptions

- **2.0 Hexagon grid** – cuts Colombo District into equal hexagons so every area is compared fairly.
- **2.1 Temporal demand index** – estimates where EV charging is needed in each period (weekday evening peak, weekday day, weekend …) from people, places and roads. (!) Period weights are assumptions until survey/pilot data.
- **2.2 Hotspots** – finds clusters of high demand; used to understand the city, not to place chargers directly.
- **2.3 AHP weights** – turns "how much more important is A than B" judgements into criteria weights and checks they are consistent. (!) Placeholder judgements until expert interviews.
- **2.4 Site scoring** – gives every real candidate site (fuel station, mall, supermarket, car park, hotel, rest area) a score from demand, grid spare capacity, road access, land and distance to existing chargers.
- **2.5 MCLP + grid** – chooses the N sites that cover the most demand within 3 km **without overloading any substation** (EV allowance = spare capacity × 0.9 × 20 % (!) assumption). Main result: ignoring the grid looks slightly better on coverage but overloads substations.

C3 – Terrain- and SoC-Aware Route and Charging-Stop Recommendation (Which route and stop?)

Datasets

Dataset	What it gives	Link	Used for	Licence
BMW i3 real driving cycles (TU Munich)	70 real trips, 1,358 km, 10 Hz: speed, elevation, battery V/I/SoC, heating	https://ieee-dataport.org/open-access/battery-and-heating-data-real-driving-cycles	**training/testing energy models**	IEEE DataPort terms
BMW i3 simulation runs (same page)	24 simulated runs (−10/0/+10 °C)	same link	physics checks only (never accuracy)	same
VED (Vehicle Energy Dataset, Michigan)	383 vehicles incl. 3 Nissan Leaf EVs + 24 PHEVs	https://github.com/gsoh/VED	backup of eVED	Apache-2.0
eVED (extended VED)	VED + map-matching, smoothed elevation, speed limits, intersections	https://bitbucket.org/datarepo/eved-dataset	**training/testing energy models (Leaf)**	Apache-2.0
Copernicus DEM GLO-30	30 m elevation for all of Sri Lanka	https://registry.opendata.aws/copernicus-dem/	road slope / climb	Copernicus licence
Open-Meteo archive	hourly temperature + humidity 2023–2025, 8 towns	https://open-meteo.com/en/docs/historical-weather-api	temperature input	CC BY 4.0
Open EV Data	battery size, AC/DC power, connectors, consumption (118 models, 2020)	https://github.com/OpenChargingCloud/open-ev-data	vehicle specs, constant kWh/km baseline	see repo licence
OpenStreetMap Sri Lanka + OSM chargers	road network (routing), 66 chargers	https://download.geofabrik.de/asia/sri-lanka.html	routes, charging stops	ODbL

Models

#	Model	Method	Baseline	Result (test trips)
3.1	Physics energy model	vehicle dynamics (rolling, drag, grade, acceleration, regen + temperature-based auxiliaries); 8 parameters fitted on training trips	constant kWh/km	BMW **91.7 %**, Leaf **87.9 %** trip accuracy (constant: 76.7 % / 65.2 %)
3.2	**Hybrid energy model**	physics + XGBoost on the remaining error	physics-only, ML-only	BMW **91.6 %** (best per segment, 5.3 Wh error); Leaf **89.2 %** (best overall, also on hills)
3.3	Routing + SoC tracking	own router on OSM major roads (643k nodes, 26,882 km, DEM elevation): Dijkstra for **shortest / fastest / energy-aware**	shortest, fastest	Colombo → Kandy 116 km, 12.7 kWh (BMW i3), < 1 s per request
3.4	Charging-stop planning	multi-stop rule scoring (position, kW, detour, connector, busyness placeholder)	nearest compatible charger	finds stops on Colombo routes; long / hill trips **not reachable with the 66 OSM chargers**

Short descriptions

- **3.1 Physics model** – calculates the energy of each ~200 m road piece from the forces on the car (tyres, air, climbing, speeding up, energy back when braking) plus heater/AC. The car's mass and drag area are not in any dataset, so they are **fitted** on training trips (results match real cars: i3 ≈ 1,550 kg, CdA 0.62 m²).

- **3.2 Hybrid model** – adds an XGBoost correction for what physics misses (driving style, traffic, temperature). About 90 % trip accuracy vs 65–77 % for a constant kWh/km. (!) Trained on flat roads (Munich, Michigan) – hill accuracy needs Sri Lankan pilot trips.
- **3.3 Router + SoC** – finds three real road routes and tracks the battery along each one against the 15 % reserve, so the driver sees where a climb uses energy and where braking downhill gives it back.
- **3.4 Stop planner** – if the battery would fall below the reserve, picks the best compatible charger before that point (one or more stops) and how long to charge. Busyness will come from C1 after merging.

Common rules behind all results

- Training and testing on **real data**, **time-based splits** (no shuffling across time); simulators only for scenarios.
- Every proposed model is compared with its **baseline** (CLAUDE.md §6).
- (!) = assumption that must be replaced when real data arrives (expert AHP judgements, POI period weights, EV share of substation capacity, CEB current loading, charger list, Sri Lankan pilot trips).
- Keep data attributions: OSM (ODbL), WorldPop / WorldCover / Open-Meteo (CC BY 4.0); Utrecht 4TU is non-commercial.